N-gram Hash Value-Embeddings and a Hash-Table Scaling Law in Fixed-Wall-Clock LLM Pretraining

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Abstract

We report rounds 11–15 (exp078–exp098) of an autonomous campaign on a fresh Karpathy autoresearch baseline (nanochat-style GPT, ∼50M base parameters, depth 8, sliding-window attention, Muon+AdamW) under a fixed 5-minute wall-clock budget on a single H200, continuing from the TOTAL_BATCH_SIZE $= 2^{18}$ + FFN 6× reference of papers 001–002 (3-seed mean val_bpb = 0.97696, single-seed $\sigma \approx 0.0015$). The rounds' contribution is a qualitatively new lever: an "Engram"-style n-gram hash value-embedding memory, in which a causal rolling polynomial hash of the last n tokens ($n \in \{2, 3\}$) indexes a size-T trainable hash table whose summed lookup is added to the value stream on the value-embedding layers. Two findings survive. First (F5), the mechanism works at essentially zero wall-clock cost: at $T = 2^{16}$ it improves val_bpb to a 3-seed mean of 0.9700 (best 0.969069; paired $\Delta \approx -0.007$, $\approx 4.6\sigma$) while retaining ∼1680 steps, with measured MFU rising from ∼28% to ∼43% because the tables add nominal FLOPs at negligible time cost. Second (F6), table size obeys a clean scaling law: val_bpb decreases approximately log-linearly in T from 2^{15} (0.97345) through 2^{20} (0.95443), ≈ -0.004 per doubling, with a 3-seed anchor at 2^{18} (mean 0.9621). The order mix interacts with T exactly as a collision account predicts: at small T bigram-only is marginally better (0.96798 at 2^{16}) and adding 4-grams hurts (0.97318), while at large T the (2,3) mix beats bigram-only (0.95443 vs. 0.95752 at 2^{20}). We adopt $(2, 3)$ at $T = 2^{20}$ as the new reference. Net campaign state: best val_bpb = 0.9544, -0.0757 (≈7.3% relative) versus the naive first-run baseline of 1.0301.

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1. Introduction

Papers 001–002 of this campaign (OPHIS, 2026a;b) established the current reference under a fixed 5-minute wall-clock budget: TOTAL_BATCH_SIZE $= 2^{18}$ tokens (F2) and FFN ratio 6× (F4), for a 3-seed mean of val_bpb = 0.97696, together with two methodological guardrails — the compile-cache confound (F1) and the single-seed noise-floor trap (F3), the latter fixing the adoption rule at 3-seed paired confirmation clearing 2σ against concurrent controls ($\sigma \approx 0.0015$). Paper 002 closed by ranking "n-gram/Engram-style memory modules" as the top follow-up, on the argument that FFN width had shown the frame to be capacity-limited (Hypothesis H2) and that memory added through a different channel than dense matmul capacity would test whether that picture generalizes.

Rounds 11–15 executed that plan. Round 11's remaining dense probes were flat or rejected — value embeddings on all layers looked promising at n=1 (0.973933) but fell to a 3-seed mean of 0.9764 and was discarded, another instance of F3 — which sharpened the motivation: the value-embedding pathway seemed worth feeding, but with content conditioned on more than the current token. Following the conditional-memory-via-lookup design of Engram (Engram, 2026), we augmented the per-layer value embeddings with bigram and trigram hash embeddings: a causal rolling hash of the last n tokens indexes a large trainable table, turning local n-gram statistics — the oldest object in language modeling (Bengio et al., 2003) — into a static memory the transformer can consult for free.

This paper documents two findings:

1. F5 (Section 3): the n-gram value-embedding memory is the campaign's third confirmed lever, worth -0.007 val_bpb at table size 2^{16} (3-seed,

≈ 4.6σ paired) at no throughput cost — step count is retained and measured MFU rises from ∼28% to ∼43%.

2. F6 (Section 4): the table size T is itself a scaling axis. val_bpb falls approximately log-linearly in T over $2^{15} \to 2^{20}$ (≈ -0.004 per doubling, cumulative -0.019), governed by a collision-vs-undertraining trade-off that also explains why the best order mix flips from bigram-only at small T to (2,3) at large T, and why 4-grams do not pay at the sizes probed.

Together the findings open a static-memory-via-lookup axis that is qualitatively unlike every previous lever: it adds parameters without adding wall-clock cost, so under this frame it is, to first order, free capacity.

2. Method: N-gram Hash Value-Embeddings

Base mechanism. The baseline architecture already routes a per-layer value embedding — a learned per-token vector of dimension d_{kv} — into the attention value stream on designated value-embedding layers. This pathway conditions values on the identity of the current token only.

N-gram lookup. For each order $n \in \{2, 3\}$ we allocate a hash table $E_n \in \mathbb{R}^{T \times d_{kv}}$. At position t, a causal rolling polynomial hash of the last n token ids,

$$h_n(t) \;=\; \Big(\sum_{j=0}^{n-1} c_j\, x_{t-j}\Big) \bmod T,$$

with fixed odd multipliers c_j, produces an index into E_n; the retrieved rows are summed over orders and added to the existing value embedding on the value-embedding layers. The hash is causal (only tokens $\leq t$ enter $h_n(t)$), rolling (computed for all positions in one vectorized pass over the shifted token tensor), and deterministic — the same n-gram always maps to the same slot, so the table is a static conditional memory in the sense of Engram (2026): content addressed by local context, retrieved in $O(1)$, with collisions resolved only implicitly by training.

Implementation. The hash is computed with compile-safe integer tensor ops (shifts, multiplies, modulo), so the mechanism fuses into the existing torch.compile graph without graph breaks — critical under a wall-clock frame where recompilation or eager fallback would eat the budget (cf. F1 (OPHIS, 2026a)). Lookup is a standard embedding gather. The tables are registered in the AdamW embedding parameter group, inheriting the embedding learning rate and schedule; Muon never sees them. Memory cost is the tables themselves ($\sum_n T \cdot d_{kv}$ parameters); at $T = 2^{20}$ observed VRAM grows from 50.8 GB (reference) to 57.9 GB. This is the hash-embedding trick of Svenstrup et al. (2017) applied not to input words but to n-gram context keys feeding the value stream.

Frame and gate. Unchanged from papers 001–002: 5-minute wall-clock on one H200, metric val_bpb (lower is better), steps completed an outcome rather than a control, single-seed runs screening-only, adoption requiring 3-seed paired confirmation at 2σ against concurrent controls ($\sigma \approx 0.0015$). All comparisons in this paper are against the paper-002 reference (3-seed mean 0.97696).

3. The Memory Lever (Finding F5)

Round 12's pilot ran (2,3)-gram tables at $T = 2^{16}$: val_bpb = 0.969687, a single-seed -0.0073 against the reference — the largest screening delta of the campaign, nearly 5σ, far past the F3 pilot bar. Round 13 confirmed it: seeds gave 0.969687/0.971339/0.969069, mean **0.9700**, improving on all three concurrent reference pairs with mean paired $\Delta \approx -0.0069$ ($\approx 4.6\sigma$). Adopted — the campaign's third confirmed lever and third milestone.

The systems profile is the notable part. Every previous capacity lever paid wall-clock: FFN 6× surrendered ∼14% of its steps to win (OPHIS, 2026b). The n-gram memory pays nothing measurable: runs retain ∼1680 steps (1699/1695/1721 across the confirmation seeds, versus ∼1720 for the reference), and measured MFU rises from ∼28% to ∼43% — the tables add nominal model FLOPs while the added step time (an integer hash and an embedding gather) is negligible, so utilization accounting improves. Under a wall-clock frame this is the best kind of lever: capacity whose throughput price is ≈ 0, sidestepping the capacity×update trade-off (H2) entirely rather than optimizing along it.

4. A Hash-Table Scaling Law (Finding F6)

With the mechanism confirmed, rounds 13–15 swept its design space: hash-table size T, and which orders to include. We state the governing hypothesis:

> Hypothesis H3 (memory-capacity-limited regime). In the 5-minute frame the model is memory-capacity-limited on local n-gram statistics: adding a large-table n-gram

lookup monotonically lowers val_bpb as T grows — collisions fall, so more distinct n-grams get dedicated, uncorrupted slots — until the hash table is undertrained, i.e. until slots are visited too rarely within the budget for their rows to learn useful content.

Confidence: 0.80.

The table-size axis. Single-seed screening at fixed order mix (2,3) gives a strikingly clean ladder: $T = 2^{15}$: 0.97345; 2^{16}: 0.96969; 2^{17}: 0.96705; 2^{18}: 0.96129; 2^{19}: 0.95802; 2^{20}: **0.95443**. Every doubling helps, by ≈ 0.004 val_bpb on average (range 0.0026–0.0058), with no flattening through 2^{20}. Two rungs are multi-seed: 2^{16} (mean 0.9700, Section 3) and 2^{18} (seeds 0.961288/0.961945/0.963115, mean **0.9621**), and the $2^{16} \to 2^{18}$ gap of -0.0079 between 3-seed means is itself $> 5\sigma$ — the ladder is not a chain of noise draws. Round 15 (in progress) is pushing $2^{21}/2^{22}$ to find the undertraining turnover H3 predicts; $(2,3)$ at $T = 2^{20}$ is adopted as the new reference.

Order mix interacts with T as collisions predict. At $T = 2^{16}$, bigram-only is marginally better than (2,3) (0.96798 vs. 0.96969), and adding 4-grams is clearly worse (0.97318): the trigram/4-gram key spaces are so much larger than 2^{16} that their lookups are mostly collision noise, contaminating the shared slot budget. At $T = 2^{20}$ the ranking flips: (2,3) beats bigram-only (0.95443 vs. 0.95752), because with collisions rare the trigram table finally stores usable conditional content, and bigram-only saturates (0.96441 $\to$ 0.96018 $\to$ 0.95752 over $2^{18} \to 2^{20}$, a visibly flattening curve versus (2,3)'s 0.96129 $\to$ 0.95802 $\to$ 0.95443). Higher orders are worth including exactly when the table is large enough to disambiguate them — a collision-vs-capacity account consistent with the hash-embedding analysis of Svenstrup et al. (2017) and with Engram's finding that lookup memory value scales with table capacity (Engram, 2026). The 4-gram flank (0.97318 at 2^{16}) remains untested at large T; H3 predicts it too would eventually pay, at still larger tables than trigrams require.

5. Results

Table 1 summarizes rounds 11–15. Δval_bpb is reported against the paper-002 reference mean 0.97696 (TOTAL_BATCH 2^{18}, FFN 6×, 3-seed); negative is better. Data: campaign_log.jsonl, exp078–exp098.

Three features stand out. First, every n-gram row in the table beats the reference — the first time in the campaign a whole mechanism family, rather than a single point in a sweep, has been uniformly positive. Second, the (2,3) table-size column is monotone across six sizes spanning 32× in T, with 3-seed anchors at 2^{16} and 2^{18} whose gap alone clears the gate several times over. Third, the bigram-only and (2,3) columns cross between 2^{16} and 2^{18}, exactly the signature the collision account of Section 4 requires — a qualitative prediction, not a fitted one.

6. Threats to Validity

1. Single-seed rungs on the scaling ladder. Of the six (2,3) table sizes, only 2^{16} and 2^{18} are 3-seed; 2^{15}, 2^{17}, 2^{19}, and 2^{20} rest on one seed each against $\sigma \approx 0.0015$. The law is supported by monotonicity across six points and two multi-seed anchors, but any individual rung — including the adopted 2^{20} value 0.95443 — could shift by ± 0.003. Per the campaign's own F3 discipline, the 2^{20} adoption is provisional pending 3-seed confirmation in round 15.

2. The order-mix crossover is n=1 on both sides. Bigram-only vs. (2,3) differ by 0.0017 at 2^{16} and 0.0031 at 2^{20} — the latter is $\sim 2\sigma$, the former within noise. The crossover's direction matches the collision account, but its location is not pinned down.

3. Parameter and memory growth. The tables add $\sum_n T \cdot d_{kv}$ parameters; at $T = 2^{20}$ they rival the $\sim$50M dense base model, and VRAM grows from 50.8 to 57.9 GB. The campaign metric is wall-clock-fair but not parameter-fair: the headline gains partly reflect that this frame prices FLOPs and seconds, not bytes. Scaling to $2^{21}/2^{22}$ will meet VRAM limits on shared hardware before it meets the metric's limits.

4. MFU as evidence. The $\sim$28% $\to$ $\sim$43% MFU rise is an accounting consequence of added nominal FLOPs at flat step time; the load-bearing throughput fact is the retained $\sim$1680 steps, and MFU should not be read as a kernel-efficiency improvement.

5. Undertraining flank unobserved. H3 predicts a turnover where table slots become too rarely visited to train, but through 2^{20} no flattening is visible in (2,3). Until the flank is observed, "until undertrained" is a prediction, not a measurement, and the 0.80 confidence reflects this.

6. Shared hardware. As in papers 001–002, cotenant load can shift step counts between waves;

Table 1. All n-gram-memory configurations, rounds 11–15. Δval_bpb is versus the FFN 6× 3-seed reference mean 0.97696 (negative = better). Single-seed rows are screening pilots; only $n \geq 3$ rows are eligible for the 2σ keep gate. All n-gram runs retain ∼1650–1720 steps.

Config	val_bpb	Δ vs. reference	Seeds	Verdict
Reference: TOTAL_BATCH 2^{18}, FFN 6×	0.97696	—	3	reference (papers 001–002)
VE on all layers (round 11)	0.9764 (best 0.973933)	−0.0006	3	reject — pilot was noise (F3)
n-gram (2,3), $T = 2^{16}$ (pilot, exp078)	0.969687	−0.0073	1	big win; confirm
n-gram (2,3), $T = 2^{16}$ (confirmation)	0.9700 (best 0.969069)	−0.0069 (≈ 4.6σ paired)	3	keep — adopted (F5)
n-gram bigram-only, $T = 2^{16}$	0.967981	−0.0090	1	≈ (2,3); slightly better at sma
n-gram (2,3,4), $T = 2^{16}$	0.973176	−0.0038	1	4-gram over-sparse, discard
n-gram (2,3), $T = 2^{15}$	0.973448	−0.0035	1	smaller table worse, discard
n-gram (2,3), $T = 2^{17}$	0.967048	−0.0099	1	scaling rung (F6)
n-gram (2,3), $T = 2^{18}$	0.9621 (best 0.961288)	−0.0148	3	keep — scaling anchor (F6)
n-gram (2,3), $T = 2^{19}$	0.958016	−0.0189	1	scaling rung (F6)
n-gram (2,3), $T = 2^{20}$	0.954428	−0.0225	1	best — adopted as reference
n-gram bigram-only, $T = 2^{18}$	0.964408	−0.0126	1	worse than (2,3) at large T
n-gram bigram-only, $T = 2^{19}$	0.960181	−0.0168	1	worse than (2,3)
n-gram bigram-only, $T = 2^{20}$	0.957518	−0.0194	1	worse than (2,3); flattening

concurrent controls mitigate but do not eliminate this.

7. Conclusion and Future Work

Rounds 11–15 opened and then scaled a new axis. F5: an Engram-style n-gram hash value-embedding memory — a causal rolling polynomial hash of the last 2–3 tokens indexing trainable tables added to the value stream — improves val_bpb by a confirmed paired −0.007 at $T = 2^{16}$ (≈ 4.6σ, 3-seed mean 0.9700) at zero wall-clock cost, with ∼1680 steps retained and MFU rising from ∼28% to ∼43%. F6: table size is a scaling law, ≈ −0.004 val_bpb per doubling from 2^{15} to 2^{20} with 3-seed anchors at 2^{16} (0.9700) and 2^{18} (0.9621), a collision-governed order-mix crossover (bigram-only best at small T; (2,3) best at large T; 4-grams over-sparse at the sizes probed), and no turnover yet at 2^{20} (0.95443, adopted as reference). Confidence in H3 is 0.80: the monotone six-point ladder with two multi-seed anchors makes the memory-capacity-limited reading hard to dismiss, but four rungs are single-seed and the predicted under-training flank has not yet been observed.

Net campaign state after fifteen rounds: best val_bpb = 0.9544, an improvement of −0.0757 (≈7.3% relative) versus the naive first-run baseline of 1.0301, with three adopted levers (TOTAL_BATCH 2^{18}; FFN 6×; n-gram (2,3) memory at $T = 2^{20}$) and the noise-floor discipline of F3 enforcing every adoption.

Ranked next experiments: (1) continue table scaling to $2^{21}/2^{22}$ until the law saturates or VRAM binds, with 3-seed confirmation of the terminal size (round 15, in progress); (2) per-order table sizing — the crossover analysis implies bigrams need fewer slots than trigrams, so splitting the parameter budget unevenly ($T_2 < T_3$) should dominate the shared-size design; (3) revisit 4-grams at $T \geq 2^{20}$, where H3 predicts they begin to pay; (4) re-check the FFN-ratio and batch-size optima under the new reference, since a large static memory may relieve the FFN of n-gram storage duty (Geva et al., 2021) and shift the capacity×update balance of H2.

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